

Once
you stop
learning,
you start
dying.

—Albert Einstein

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TABLE 2
CONNECTIONS BETWEEN
GESTURE SENSOR AND
ARDUINO MEGA

Gesture Sensor (S1)	Arduino Mega (MOD1)
VCC	5V
GND	GND
SDA	SDA
SCL	SCL

TABLE 3
CONNECTIONS BETWEEN
4-CHANNEL RELAY MODULE
AND ARDUINO MEGA

4-Channel Relay Module (RL1)	Arduino Mega (MOD1)
VCC	5V
GND	GND
IN 1	D2
IN 2	D3
IN 3	D4
IN 4	D5

electronicsforu.com and uploading it to the microcontroller using the Arduino IDE. Ensure the correct board and port are selected before uploading.

The system begins by initialising the gesture sensor and relay module, preparing them for operation. Once powered on, it enters a continuous loop to monitor gesture inputs. When a gesture is detected, the microcontroller interprets the motion and performs the corresponding action:

- **Left/right gestures.** Navigate between appliances by incrementing or decrementing the relay counter
- **Down gesture.** Turns ON the currently selected appliance
- **Up gesture.** Turns OFF the selected appliance

The relay is connected to the normally closed (NC) and common (COM) terminals, allowing a 12V DC fan to serve as an indicator when the relay is activated. The system maintains circular navigation logic,

```
#define RELAY_1 2
#define RELAY_2 3
#define RELAY_3 4
#define RELAY_4 5
```

Fig. 3: Code snippet – Define relay pin

```
if (gestures & GESTURE_RIGHT) {
    x = (x < 4) ? x + 1 : 1;
}
if (gestures & GESTURE_LEFT) {
    x = (x > 1) ? x - 1 : 4;
}
```

Fig. 4: Code snippet – Check and process gesture inputs

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resetting the counter to 1 after 4 (or to 4 if it drops below 1), allowing smooth and uninterrupted appliance selection.

This integration of gesture detection with relay control results in an intuitive and responsive home automation system that enhances the user experience through hands-free operation.

To test the system, power on the device and wave a hand above the gesture sensor. Perform left/right or up/down gestures to trigger the corresponding functions. The sensor will detect the gestures, and the code will control the respective relay. Any load—such as a motor, fan, or bulb—may be connected to the relay to verify proper operation. **EFY**



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